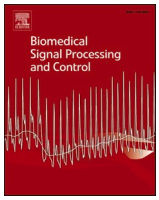
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Optimum target range bin selection method for time-frequency analysis to detect falls using wideband radar and a lightweight network

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ABSTRACT

Objective: Wideband bio-radars have been applied to fall behaviour recognition research in the last few years. The optimum target range bin is that the time Doppler spectrogram image of signals of the range time matrix in this range bin is best for the deep learning network to distinguish between falling and non-falling. Thus, selecting an appropriate target range bin for time-frequency analysis is very important for effective fall detection of wideband radar.

Methods: A K-band wideband frequency modulated continuous wave radar is applied to build a fall detection database of 36 subjects in two scenes. The radar data are pre-processed to obtain range-Doppler spectrograms where the optimum target range bin is selected for the subsequent time-frequency analysis. The original, threshold, and binary time Doppler spectrograms are compared using different target range bin optimum target range bin selection methods. To determine the effectiveness of the proposed algorithm in fall recognition, a MobileNetV3-Small architecture is implemented for fall detection by automatically extracting features and classifying them.

Results: The proposed method distinguishes falls from non-falls with a 98.93% classification accuracy with 5-fold cross validation, which is superior to the existing maximum variance method that is applied to the dataset using three kinds of time Doppler spectrograms. The test results in another scene show that our proposed method is robust enough to detect falls successfully for unseen subjects in unseen situations.

Conclusions: The proposed optimum target range bin selection method can effectively and robustly detect falls when using wideband radar and a lightweight network.

1. Introduction

Due to the effect of a longer life expectancy, the process of ageing in the global population has accelerated. The World Health Organization (WHO) projected that by 2050, nearly 17 percent of the world's population (1.6 billion) will be elderly (age 65 and above) [1]. According to a WHO report, the probability of falling is approximately 28%–35% for elderly individuals over 65 years old and 32%–42% for elderly individuals over 70 years old [2]. Falling is the main cause of death due to accidental injury for elderly individuals over 79 years old [3]. Elderly individuals who have experienced falls are more likely to die than those who have never fallen, with an odds ratio of 5.11 (95% confidence interval: 1.84–14.17, $p = 0.002$) [2]. Falls seriously threaten the physical and mental health of the elderly and even cause death, placing a greater burden on the family and society. Noury et al. showed that if elderly individuals could obtain timely assistance after falling, they could effectively reduce the risk of death by 80% and the risk of long-term hospitalisation by 26% [4]. Therefore, if we can detect falls and notify someone in a timely and accurate manner for the elderly (especially empty nesters who live alone), the elderly can be provided with fast and effective assistance to minimise the negative impact of falls

In the last few years, radar has been applied to research on human behaviour recognition [5], especially for fall recognition [6,7] because its performance is unaffected by changes in light, sound and temperature and it has good user privacy protection. Compared with wearable devices, such as inertial measurement unit (IMU) sensors and environment sensing devices, including cameras, the cost of a radar is relatively high [8]. However, with the maturity of radar chip technology and mass production of bio-radars, the cost of a radar as one of the smart home sensors will be further reduced and widely used. Moreover, the number of subjects in the existing studies is relatively small, where most studies have two to three subjects [6], and the largest number of subjects is 20 [9], most of the experiments are carried out in a single scene or even a microwave anechoic chamber [10,11].

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In previous studies, continuous wave (CW) Doppler radars were mostly implemented for fall detection. There is little or no range resolution due to the narrow bandwidth of the CW Doppler radar [12]. For this kind of radar, the signal processing methods of network front-ends are mainly time–frequency analysis, such as short time Fourier transform (STFT) and wavelet transform [10]. The time–frequency analysis results reflect the variation law of different frequency components in the radar echo signal with time. Next, the extracted feature indices from the time–frequency analysis results are put into machine learning models, or the time–frequency analysis results are directly sent into deep learning networks to distinguish different action and behaviour types [12].

With the development of microelectronics and chip technology, several wideband continuous wave bio-radars have emerged in the last decade [13,14], which have a high resolution of range and velocity at the same time. Compared with the CW Doppler radar, wideband bio-radars can provide more multidimensional information for human behaviour recognition to improve recognition accuracy. Similarly, the extracted feature indices from the multidimensional information are put into machine learning models, or the multidimensional spectrograms are directly sent into deep learning networks to determine whether a fall occurred [15].

Compared with the CW Doppler radar, the main difference in the data processing of the wideband CW bio-radar in the network front-end radar is the multiple range resolution cells of the range profile [14]. The radar signal in each range bin changes with the slow time. The time Doppler spectrograms for each range bin can be obtained after time–frequency analysis. As overlapping sampling is employed using a sliding window, a time–frequency analysis needs more calculation than a Fourier transform. Thus, it is impossible to implement time–frequency analysis on the time sequence signal for all range bins, especially for an online real-time processing situation. Therefore, selecting an appropriate target range bin for further time–frequency analysis is very important for wideband bio-radar data processing. Hamidreza et al. proposed the maximum variance method (MVM) [14], which was the range bin that had the maximum variance of the range time matrix and was chosen as the optimum range bin. STFT was carried out on the time sequence signal in a period of the optimum range bin to obtain a spectrogram, which was converted into a binary image and fed into a convolutional neural network [14]. The binary time–frequency signature of each activity was obtained by applying the k-means clustering algorithm to the time–frequency representations. The 5-fold cross validation in one scene achieved 93.54% accuracy using data from 10 subjects [14]. The accuracy decreased to 89.12% in another test scene [14]. As there may have been interferences in the range time spectrograms, which would lead to high variance of the range time matrix, the time Doppler spectrogram signature obtained from MVM and STFT may not have been able to truly reflect the real change process of the target frequency components with time.

Considering that range Doppler spectrograms can better reflect the span of target velocity than range time spectrograms, this paper proposed an optimal target range bin selection method based on the range Doppler spectrogram. In the last decade, deep learning networks tended to have deeper and more complicated structures to achieve higher accuracy. However, complex networks could not be executed online due to the limited computing resources in many practical online applications. Thus, some scholars have developed lightweight deep learning networks that could be easily matched to the design requirements for mobile and embedded vision applications [16]. A class of efficient models called MobileNets describes lightweight deep neural networks that efficiently trade off latency and accuracy. The latest MobileNet version, i.e., MobileNetV3-samll [17]' is implemented to automatically extract features from the time Doppler spectrograms for classification. The framework of this paper is illustrated in Fig. 1. The wideband radar echoes are pre-processed by the optimum target range bin selection method in Section 2 to obtain the time Doppler spectrograms. The images of the original, threshold, and binary time Doppler spectrograms

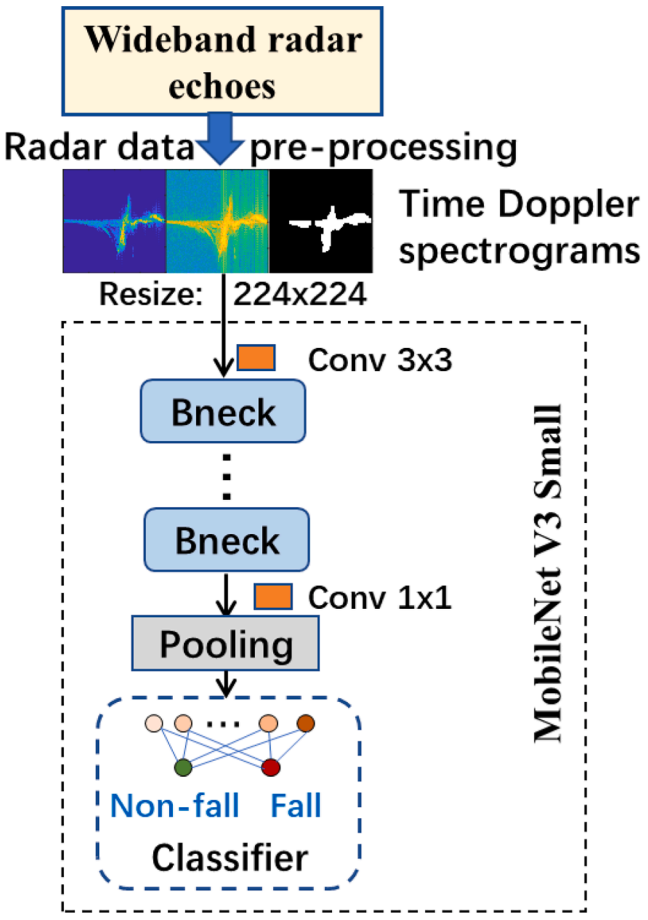


Fig. 1. The framework of this paper using a wideband radar and a lightweight network. The images of the original, threshold, and binary time Doppler spectrograms are separately input into the deep learning network. Bneck represents a bottleneck block with residual and squeeze-and-excite in MobileNet V3. Conv is the convolution operator. A fully connected layer with a softmax operation outputs the prediction as a classifier.

are separately input into the deep learning network. After resizing the time Doppler spectrogram images, the resized images are sent to the MobileNetV3-samll described in Section 3 to automatically extract features and classify falls or non-falls. Additionally, the proposed method is compared with the MVM using the original, threshold, and binary time Doppler spectrograms. To make the comparison results convincing, our team establishes a large radar noncontact fall detection database with comprehensive activity types, including 20 daily life activities and 10 fall activities of 36 subjects in two scenes. The database is divided into a training set, verification set and test set. The subjects in the three sets are different, and the scenes of the training and verification sets and the test set are different. To make the results comparable, all random number seeds are fixed when the deep learning network operates.

The paper is organised as follows: In Section 2, the proposed optimum target range bin selection method for time–frequency analysis to detect falls is described in detail, including the procedures for optimum target range selection, the calculation of the range-Doppler and the time Doppler spectrograms. In Section 3, MobileNetV3-Small, the implementation of the neural network and the performance evaluation metrics are described. The experiment and results are presented in Section 4. Finally, the conclusion is drawn in Section 5.

2. Optimum target range bin selection method based on the range-doppler spectrogram

Fig. 2 presents the flowchart of the proposed optimum target range bin selection method based on the range-Doppler spectrogram. The input is the echoes of the wideband radar. Section 2.1 describes how to obtain the range time matrix, the range time spectrogram matrix after moving target indication (MTI) and the range-Doppler spectrogram matrix. The proposed optimum target range bin selection method is explained in Section 2.2. The method of the calculation of the time Doppler spectrogram is given in Section 2.3. The output is the time Doppler spectrograms. Details of each step are described below.

2.1. Calculation of the Range-Doppler spectrogram

There are two main kinds of waveforms for wideband CW bio-radars: stepped frequency (SF) CW and frequency modulated (FM) CW. Using each period of the radar echo after de-ramping [18] and range compression (i.e., the Fourier transform) [19], a row vector containing M range bins is obtained, as shown in Fig. 3(a). De-ramping (or mixing) is when the received radar echo signal is multiplied by the replica and the conjugation of the transmitted signal, and then passed through a designed low pass filter, which can dramatically reduce the sampling rate for an SFCM and FMCM radar. The output of the de-ramping or mixing process is the beat signal. Range compression applies the Fourier transform to the beat signal to provide the different range bins, which is referred to as fast time. If N period signals are processed above and arranged line-by-line, we can obtain an $N \times M$ range time matrix RT seen in Fig. 3(b). The normalisation operator is that each element in matrix RT is divided by the maximum absolute value of RT. The operator dB is as follows:

$$dB(\boldsymbol{RT}) = 20*\log_{10}(abs(\boldsymbol{RT})) \quad (1)$$

The matrix of $\boldsymbol{RT}$ after normalisation and the dB operator is defined as the range time spectrogram matrix, which contains a large amount of static environment clutter. **MTI** is performed in the slow time direction to suppress static clutter [20]. The range time spectrogram matrix after MTI is defined as $\boldsymbol{RT}_{\mathrm{MTI}}$.

Next, a Fourier transform (FT) is performed along the slow time axis of the $\boldsymbol{RT}_{\mathrm{MTI}}$ in every range bin. The pink column in Fig. 3(b) indicates the position of the signals along the slow time axis of the $\boldsymbol{RT}_{\mathrm{MTI}}$ in some range bin for the FT. After FT in every range bin, the $N_{fft} \times M$ range Doppler matrix $\boldsymbol{RD}$ is obtained, where N_{fft} is the number of fast Fourier transform (FFT) points and M is the number of range bins. The range Doppler matrix $\boldsymbol{RD}$ after zero-frequency component shift (moving the zero frequency components to the centre of the spectrum), normalisation and dB operator is defined as the range Doppler spectrogram matrix, as seen in Fig. 4 (a). Although the static clutter is removed, the range Doppler spectrogram still contains system white noise or some strong system jitters along the frequency axis separately presented in Fig. 4(a) and (b). The parts in the red circles need to be further suppressed to calculate the optimal range bin.

We ensure that the high-energy part of the range time matrix after the MTI of the collected moving targets is roughly placed in the centre of the data duration when intercepting data, which can be realised by programming. Then, the moving target is relatively concentrated near the centre of the range-Doppler spectrogram, and thus, there are almost no targets in the four corners of the range-Doppler spectrogram (seen in Fig. 4).

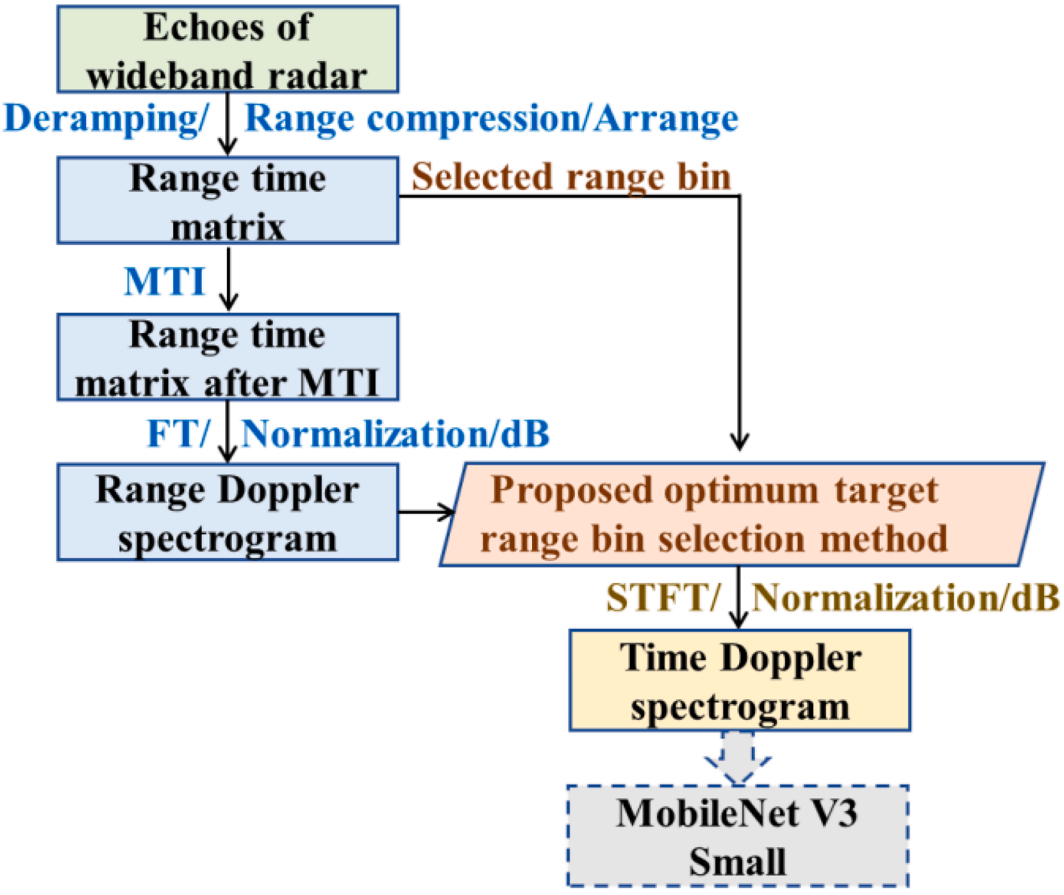


Fig. 2. A flowchart of the proposed optimum target range bin selection method based on the range-Doppler spectrogram.

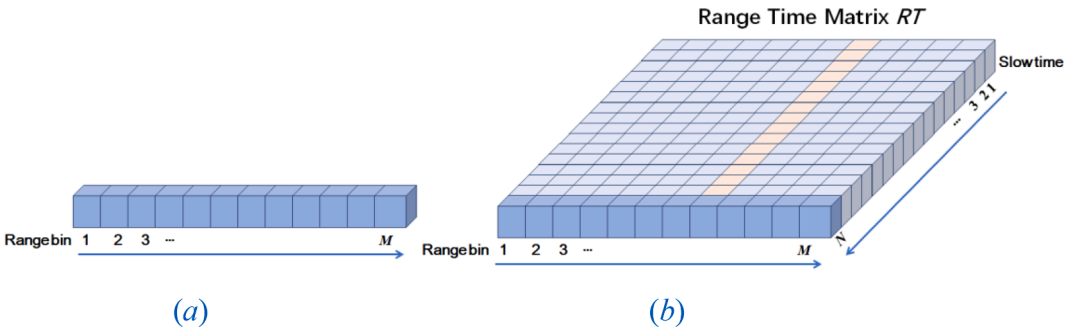


Fig. 3. Range time matrix of a wideband radar: (*a*) one dimension range bin vector of one period. (*b*) Arranged multirange bin vectors of multi-periods into a range time matrix *RT*.

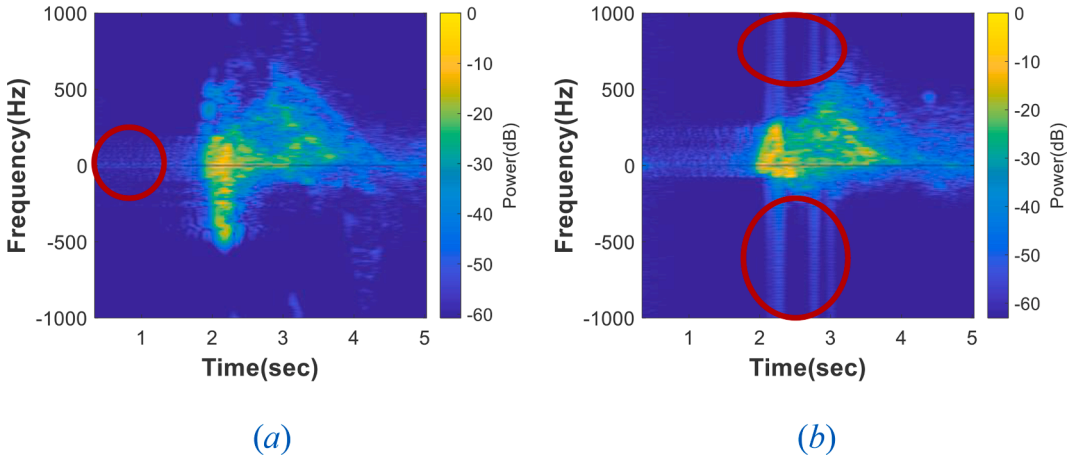


Fig. 4. Range Doppler spectrograms: (*a*) Fainting while standing up from a chair; (*b*) Falling forward. The parts in the red circles are white noise and some strong system jitters separately in (*a*) and (*b*).

2.2. Optimum target range bin selection

The optimum target range bin is that in this range bin, the time Doppler spectrogram image of signals of the range time matrix is best for the deep learning network to distinguish between falling and non-falling. The proposed optimum range bin selection method based on the range-Doppler spectrogram is as follows.

Step 1: We calculate and denote the median value of the range-Doppler spectrogram as *a*. The transformation factor is denoted by *c,* which is given an initial value, i.e., $c = 1$.

Step 2: To suppress the noise of the range-Doppler spectrogram, a binary mask is constructed. We calculate the threshold $Th = a \times c$. In the range-Doppler spectrogram matrix, an element less than the threshold is set to 0, and an element larger than or equal to the threshold is set to 1. The initial binary mask matrix is obtained, which is represented as Mask.

Step 3: As there are almost no targets in the four corners of the range-Doppler spectrogram in the proposed method, the mask is modified according to these characteristics to suppress more noise. We count the number of nonzero elements at the four corners of Mask to calculate the noise point number, which are denoted as Cn_1, Cn_2, Cn_3, and Cn_4 without order. The size of the four corner matrices is usually less than 1/50 of the size of the range-Doppler spectrogram matrix, where a parameter b_1 (=1/50) should be adjusted according to the noise figure of the radar system equipment. The rows and the columns of the range-Doppler spectrogram are N_{fft} and M, respectively. We suppose the size of the four corner matrices is$C_{N_{fft}}\mathrm{C}_{\mathrm{N_{fft}}} \times C_M$, where$C_{N_{fft}} \le b_1 N_{fft}$ and $C_M \le b_1 M$. We calculate the sum of the nonzero element numbers of the four corner matrices, denoted as CNS (=$Cn_1 + Cn_2 + Cn_3 + Cn_4$), which reflects the system noise.

Step 4: To suppress the strong system jitter along the frequency axis, two strips at the top and bottom of the matrix mask are taken to calculate the noise point number, both with size$C_{N_{fft}} \times M$, where$C_{N_{fft}} \le b_2 N_{fft}$ (parameter b_2 (=1/50) is adjustable based on the noise figure of the radar system equipment). The number of nonzero elements at the two strips are counted and represented as Cn_{up} and Cn_{down}. To retain the target with a large Doppler frequency, the smaller one is chosen and denoted as CNM, which can reflect the system jitter. That is, CNM = min(Cn_{up}, Cn_{down}).

Step 5: We adjust the transformation factor c according to the thresholds, CNS and CNM. We set the threshold $Th1 = b_3 \times C_{N_{fft}} \times C_M$ and the threshold $Th2 = b_4$, where parameters b_3(=20) and b_4(=1) can be fine-tuned according to the specific system noise. When CNS $\ge Th1$ and CNM $\ge Th2$, let $c = c\text{-}\triangle c$ update c, where $\triangle$c is the step size with a default value of 0.1.

Step 6: We repeat Steps 2 to 5 until CNS $< Th1$ and CNM $< Th2$ to acquire the final threshold $Th = a \times$ c. In the range-Doppler spectrogram matrix, elements less than or equal to the threshold are set to *Th*, and elements larger than the threshold remain unchanged. The updated range-Doppler spectrogram matrix is denoted as ***RD***', still with the dimension $N_{fft} \times M$. The system noise and the jitter in all the updated range-Doppler spectrogram matrices are suppressed well when using the chosen parameters, which are repeatedly optimised.

Step 7: Because the reflection of close targets is large, if ***RD***' is summed directly along the column, the optimum range bin will deviate from the truth. Therefore, partial components of ***RD***' with large amplitude shall be limited first. We sort all the elements of ***RD***'

in ascending order to form a vector ***X***. The percentile of the elements in the vector ***X*** for the percentage p in the interval [0,100] is calculated and denoted as x. In matrix ***RD***', an element less than or equal to $\times$ remains unchanged, and an element larger than $\times$ is set to x. The percentage p equal to 98 is the best choice in this study for all the range-Doppler spectrogram matrices.

Step 8: Next, we sum up the elements of the updated ***RD***' along the column (Doppler frequency axis) to obtain a row vector ***SRD*** with size $1 \times M$. The range bin corresponding to the maximum value of the ***SRD*** is selected as the optimum range bin.

During the selection of the optimum range bin, the reason why we choose the range bin with the maximum value of the ***SRD*** instead of the range bin with the largest Doppler value is that it can suppress the false components carried by the artefacts of the multipath interference of moving targets in the environment, while considering the stable Doppler maximum components, which contain complete and proper information of the frequency changing activity with time.

2.3. Calculation of the time Doppler spectrograms

A time–frequency analysis is performed by applying the STFT to the time series of the range time matrix RT on the optimum range bin. The number of frequency bins of the STFT is N_{fft}. A unit energy Hamming window with a length of 128 is chosen as the frequency smoothing window of the STFT. The normalised time Doppler spectrogram is called the original time Doppler spectrogram.

We denote the median value of the time Doppler spectrogram as a and set a threshold of $c'a$, where c' is a parameter between 0 and 1. For the original time Doppler spectrogram matrix, an element less than or equal to $c'a$ is set to $c'a$, and an element larger than $c'a$ remains unchanged. The updated time Doppler spectrogram matrix is defined as the threshold time Doppler spectrogram. In previous studies [6,14,21], the threshold was usually set to a fixed number in the normalised time Doppler spectrograms. However, for spectrograms of different activities, the fixed threshold is not rational. Thus, the median value (usually approximately -100 dB in this study) multiplied by a factor c' is proposed in this paper for the threshold time Doppler spectrogram.

Moreover, the binary time Doppler spectrogram matrix obtained by applying k-mean clustering, median filtering and a morphological opening operation [14] preserves the shape of the energy content of the activity. The images of the binary time Doppler spectrogram are fed into the convolutional neural network. Because the time dimension of the time Doppler spectrogram matrix is large, the time Doppler spectrogram matrix before k-mean clustering is resized to 224×224 to reduce the amount of computation.

Figs. 5 to 7 show the original, threshold and binary time Doppler spectrograms of walking and falling, respectively, where c' equals 0.75 for the threshold time Doppler spectrograms. As these images of the threshold time Doppler spectrograms look very clean and are almost noise free, c' equals 0.75 in the following experiment. The influence of the different values of parameter c' on the classification results will be discussed in Section 4.2. Graphs (a) and (c) in Figs. 5 to 7 are obtained using the proposed method. Graphs (b) and (d) in Figs. 5 to 7 are acquired using the MVM. Compared with the spectrograms using the MVM, the time Doppler spectrograms using the proposed method can reflect more detailed information on activities, which is beneficial for recognising the next action. The patterns in the spectrograms of walk

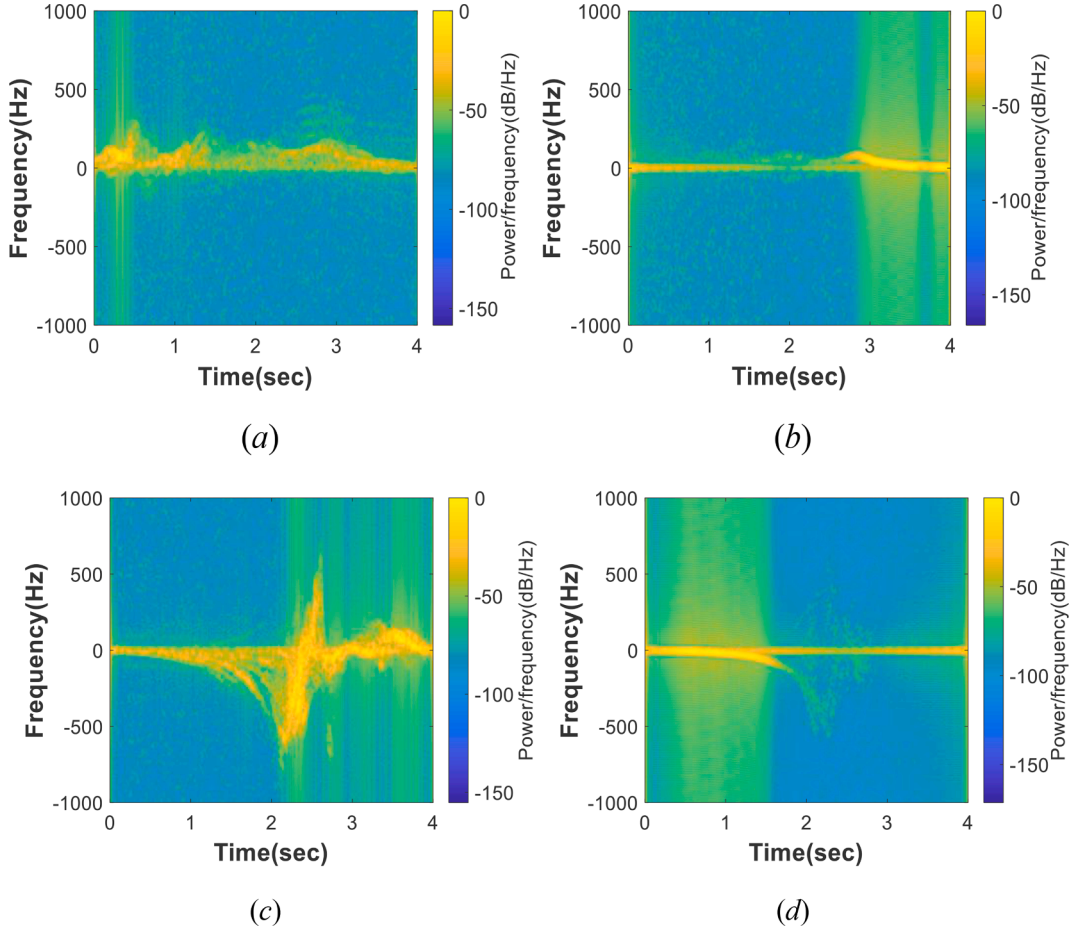


Fig. 5. Original time Doppler spectrograms: (*a*) walking towards a radar using the proposed method, (*b*) walking towards a radar using the MVM, (*c*) slipping backwards using the proposed method, (*d*) slipping backwards using the MVM.

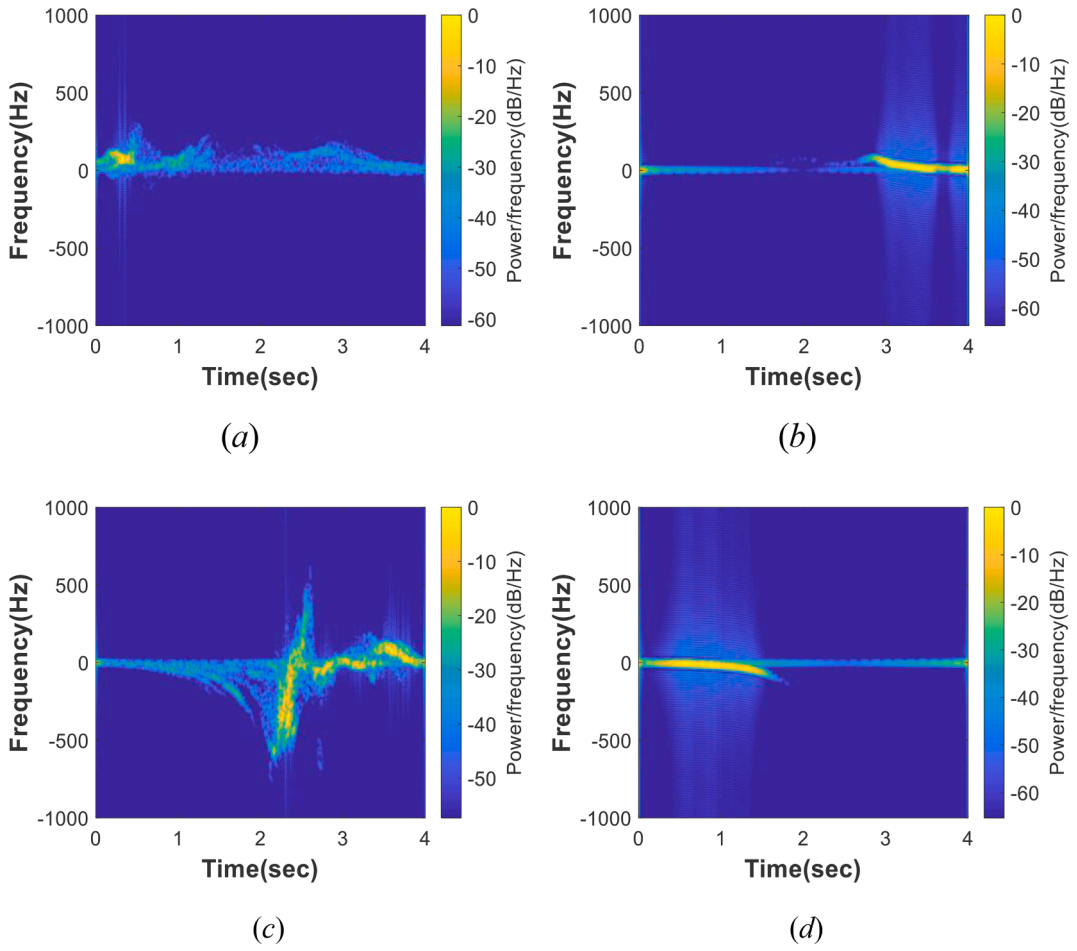


Fig. 6. Threshold time Doppler spectrograms with *c'* equal to 0.75: (*a*) walking towards a radar using the proposed method, (*b*) walking towards a radar using the MVM, (*c*) slipping backwards using the proposed method, (*d*) slipping backwards using the MVM.

and slip are different. The spectrograms of slip have larger frequency components than the walk spectrograms. There are positive and negative frequencies in the original, threshold and the binary time Doppler spectrograms. However, the time Doppler spectrograms in [14] only present the positive frequency parts.

3. Lightweight network

3.1. MobileNetV3-Small

Since 2017, the Google team has presented a series of MobileNets focusing on a lightweight convolutional neural network (CNN) in mobile or embedded devices [16]. Compared with a traditional CNN, it can greatly reduce the model parameters and the amount of computation on the premise of a small reduction in accuracy [16]. MobileNetV3-Small is one of the latest architectural designs [17] and includes 12 bottleneck residual layers and 4 convolution layers [17]. The particular architecture of the MobileNetV3-Small used in this work is listed in Table 1. The differences between the architecture used in this study and that in [17] are the input and the output size. Considering the input with only one channel, the input size in Table 1 is $224^2 \times 1$. The last output size in Table 1 is 2, as there are two categories of a fall and non-fall. Moreover, the squeeze-and-excite is applied in the residual layer, and the hard version of the swish function is designed as the activation function [17]. The specific configuration of the bottleneck residual layers in the MobileNetV3-Small architecture can be found in [17]. Batch normalisation in a deep learning network is different from the definition of normalisation in Section 2.1. The well-trained architecture of the MobileNetV3-Small can reflect the quality of the different input data.

3.2. Implementation of the neural network

The inputs of the MobileNetV3-Small include one of the original, threshold, and binary time Doppler spectrograms. For consistency, the original and the threshold Doppler spectrograms are both resized into 224×224 matrices. The MobileNetV3-Small is implemented to automatically extract features from the time–frequency representation containing the energy of activities. Fivefold cross validation and the cross-entropy cost function are employed on the training and validation datasets to optimise the hyperparameters. To compare the results using the three kinds of spectrograms, random number seeds for the CPU, GPU and dataset division in the deep learning training and testing need to be fixed.

The MobileNetV3-small hyperparameters of the training dataset are as follows: the batch size is 128, the learning rate is 0.01, the dropout regularisation factor is 0.8, and the momentum is 0.9. Finally, the size of the optimised MobileNetV3-Small model using all of the data of the training and validation datasets is 1.66 Mbytes.

3.3. Performance evaluation metrics

To evaluate the effectiveness of the trained classification model, some performance evaluation metrics are applied in this study, which mainly include the receiver operating characteristic curve (ROC), the area under the ROC curve (AUC), the sensitivity, the specificity, the accuracy, the precision and the $F_{1\text{-score}}$. The formulas are as follows:

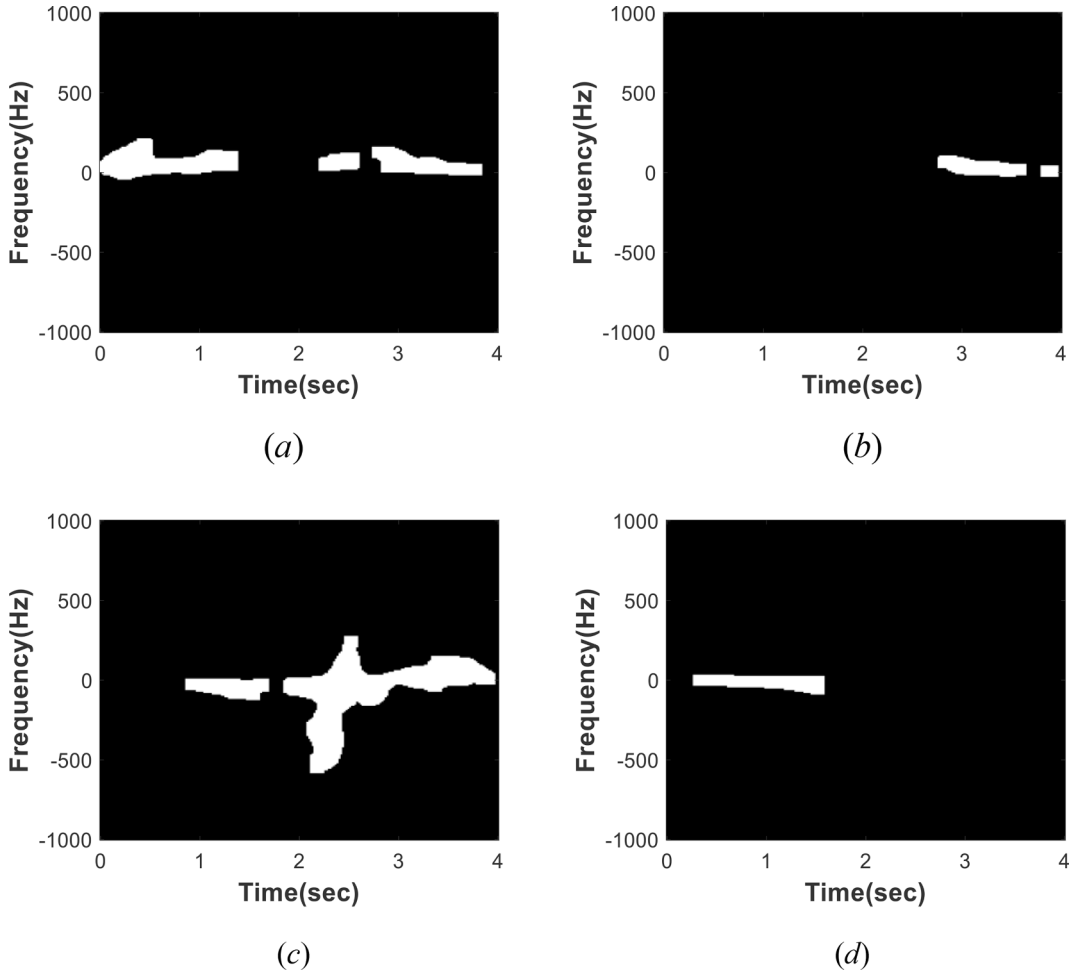


Fig. 7. Binary time Doppler spectrograms: (*a*) walking towards a radar using the proposed method, (*b*) walking towards a radar using the MVM, (*c*) slipping backwards using the proposed method, (*d*) slipping backwards using the MVM.

Table 1
Specifics of the MobileNetV3-Small architecture used in this work.

Input	Operator	Exp_size	#out
$224^2 \times 1$	conv2d, 3×3	–	16
$112^2 \times 16$	bneck, 3×3	16	16
$56^2 \times 16$	bneck, 3×3	72	24
$28^2 \times 24$	bneck, 3×3	88	24
$28^2 \times 24$	bneck, 5×5	96	40
$14^2 \times 40$	bneck, 5×5	240	40
$14^2 \times 40$	bneck, 5×5	240	40
$14^2 \times 40$	bneck, 5×5	120	48
$14^2 \times 48$	bneck, 5×5	144	48
$14^2 \times 48$	bneck, 5×5	288	96
$7^2 \times 96$	bneck, 5×5	576	96
$7^2 \times 96$	bneck, 5×5	576	96
$7^2 \times 96$	conv2d, 1×1	–	576
$7^2 \times 576$	pool, 7×7	–	–
$1^2 \times 576$	conv2d, 1×1, NBN	–	1024
$1^2 \times 1024$	conv2d, 1×1, NBN	–	2

Abbreviations: Exp_size denotes the expansion size of channels after the bottleneck; #out is the output channel number; bneck represents a bottleneck block with a residual and Squeeze-and-Excite; conv2d denotes 2-dimensional convolution; pool represents the average pooling; NBN is no batch normalisation.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Sensitivity} = \frac{TP}{P} \tag{3}$$

$$\text{Specificity} = \frac{TN}{N} = \frac{\text{TN}}{\text{N}} \tag{4}$$

$$\text{Accuracy} = \frac{TP + TN}{P + N} = \frac{\text{TP} + \text{TN}}{\text{P} + \text{N}} \tag{5}$$

$$\text{F}_{1-\text{score}} = \frac{2\text{Precisionn Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{6}$$

where P and N represent the number of fall and non-fall activities, respectively, TP is the true positive, FN is the false negative, TN is the true negative, and FP is the false positive.

4. Experiment and results

4.1. Experiment and dataset

A K-band wideband linear frequency modulated CW radar (developed by WuHan RadarDoctor Electronics and Technology CO., Ltd.) is employed in the experiment to collect the data. The central frequency of the K-band radar is 23.8 GHz and the bandwidth is 2 GHz. The pulse repetition frequency, i.e., the sampling frequency of the slow time, is

2000 Hz. A triangular linear frequency modulated waveform is adopted to maintain a stable frequency variation. The radar is mounted 2 m above the floor level.

Thirty-six subjects were recruited and approved by the ethics committee of Third Military Medical University (Army Medical University) with approval number AF/sc-03/1.0. The statistical physical information of the enrolled subjects is listed in Table 2. As the number of subjects was small ($3 < n < 50$), the Shapiro–Wilk statistical test method was used to test whether the variables obeyed a normal distribution. The results show that there was no substantial difference in the physical parameters, including height, weight and body mass index (BMI), which indicates that only age does not follow a normal distribution. Thus, the following statistical information of the age is expressed in quartiles as the median (1st quartile, 3rd quartile), while the other parameters are expressed as the mean ± standard deviation (std).

Due to ethical constraints, it is not possible to collect data on falls of the elderly in the experiment with risk. The subjects were asked to imitate the behaviour of the elderly in this experiment through 20 kinds of daily activities, such as walking, jogging, sitting down, lying down, standing up, going up and down stairs, crawling and walking on crutches, and 10 kinds of fall activities on a sponge mat, such as falling when sitting down or standing up and fainting and slipping in different directions. The description of the different activities is listed in Table 3. We collected radar data in two real room scenes without any constraints. Both rooms had many types of furniture, such as a bed, a short ladder, a desk, some iron filing cabinets, some chairs, a sponge pad and a computer, and the dimensions of the rooms were 6.40 m × 4.92 m × 2.46 m and 5.6 m × 2.47 m × 2.63 m (length × width × height). Scene 1 was composed of the data of 30 subjects (shown in Fig. 8(a)), and Scene 2 was composed of the data of 6 subjects (shown in Fig. 8(b)). The duration of each data recording was at least 4 s. In the calculation of the following spectrograms, each data segment of 4 s was intercepted. Every activity data point had a label, where 0 represented a daily life activity and 1 represented a fall activity.

To make the trained network model have better generalisability and measure the real performance of the network classification at the same time, the principle of the mutual exclusion of the samples was followed in the dataset division. The data of every subject was either in the training set, in the verification set, or in the test set. The whole dataset contains 3240 samples. Data collected in Scene 1 was further divided into training and verification sets. Data collected in Scene 2 constituted the test set.

4.2. Experimental results

The ROC curves and the AUCs of the proposed method through 5-fold cross validation are presented in Fig. 9, where the value of the AUC equals 1 due to the retention of three substantial digits and rounding. Fig. 9(a)–(c) show the results using the original, threshold, and binary time Doppler spectrograms, respectively. It can be seen from these figures that the AUCs obtained by using the original and the threshold time Doppler spectrograms are higher and have smaller standard deviations than those obtained by using the binary time Doppler spectrograms.

The cross-entropy loss and the accuracy values for the training and the validation datasets were acquired using the proposed method and the MVM, as shown in Fig. 10. When the number of epochs is larger than 15, the cross-entropy loss and accuracy values remain steady. Thus, only the results of 30 epochs are presented in Fig. 10. It can be seen from Fig. 10 that the loss using the proposed method decreases more rapidly and is smaller than that using the MVM when the model reaches a steady state. Moreover, the accuracy using the proposed method increases more rapidly and is larger than that using MVM for the training and the validation datasets.

Table 4 lists the classification metrics obtained using 5-fold cross-validation for the proposed method and the MVM. The datasets are all from Scene 1. It can be seen from this table that the proposed method outperforms the MVM using the original, threshold and binary time Doppler spectrograms. The proposed method achieves 98.926% classification accuracy, 99.222% specificity, and 98.388% $F_{1\text{-score}}$. Table 5 lists the classification metrics obtained using the test dataset in Scene 2 for the proposed method and the MVM. Even in a new scene with unseen subjects, the proposed method achieves high classification performance by providing 95.370% accuracy and 97.222% specificity. Moreover, it can be seen from these tables that the original and threshold time Doppler spectrograms outperform the binary time Doppler spectrograms using the proposed method and the MVM in the two scenes. This infers that the original and the threshold time Doppler spectrograms provide more information to understand the fundamental composition of human activities and determine whether a fall occurs during daily human activities.

To analyse the effect of parameter c' on the classification results, we compare the testing results of Scene 2 using the original time Doppler spectrograms and the threshold time Doppler spectrograms with different parameter c' values of 0.75, 0.8, 0.85, 0.9, and 0.95. As the accuracy and F-score can comprehensively reflect the classification performance, they are selected as indicators. Fig. 11 shows the accuracy and the F-score of testing in Scene 2 increasing with parameter c', which indicates that the performance worsens when the threshold value increases. Compared with the original images, there is a loss of information for the threshold and the binary approaches, regardless of how to adjust the threshold. The binary approach is further simplified based on the threshold approach due to worse performance than the threshold approach (as seen in Tables 4 and 5). In Tables 4 and 5, the results of the threshold and the mask approach are worse than those of the original images, and it can be inferred that the clutter and the noise can be suppressed automatically by the deep neural network.

Su Boyu et al. implemented a wavelet transform to extract the

Table 2
Statistical physical information of the 36 enrolled subjects.

Parameter(Unit)	Value
Age(year)	22.0(20.0,24.75)
Height(cm)	169.1 ± 8.0
Weight(kg)	59.9 ± 9.5
BMI(kg/m^2)	20.8 ± 2.1

BMI: body mass index. The statistical values are expressed as the median (1st quartile, 3rd quartile) or mean ± std.

Table 3
Types of activities performed by the 36 subjects.

Category	Motion Description
Fall	Fall forward and backwards (2 kinds)
	Faint forward and backwards (2 kinds)
	Accidentally slip while sitting down on a chair
	Faint while stand up from a chair
	Fall at ± 45°angles (2 kinds)
	Trip up the stairs
	Step down the stairs and trip
Non-fall	Walk towards or away from a radar (2 kinds)
	Walk towards and away from a radar with a pair of crutches (2 kinds)
	Walk towards or away from a radar with a crutch (2 kinds)
	Jog towards and away from a radar (2 kinds)
	Go up and down stairs (2 kinds)
	Bend down low to tie one shoe
	Tie one shoe and stand up
	Bend down low to pick up a pen
	Pick up a pen and stand up
	Swivel 180°
	Laying down on the bed
	Sit up from lying on the bed
	Crawl on a yoga mat

'2 kinds' denotes that there are two kinds of activities in one description.



Fig. 8. Two indoor scenes for radar data measurement, including different activities: (*a*) Scene 1 and (*b*) Scene 2.

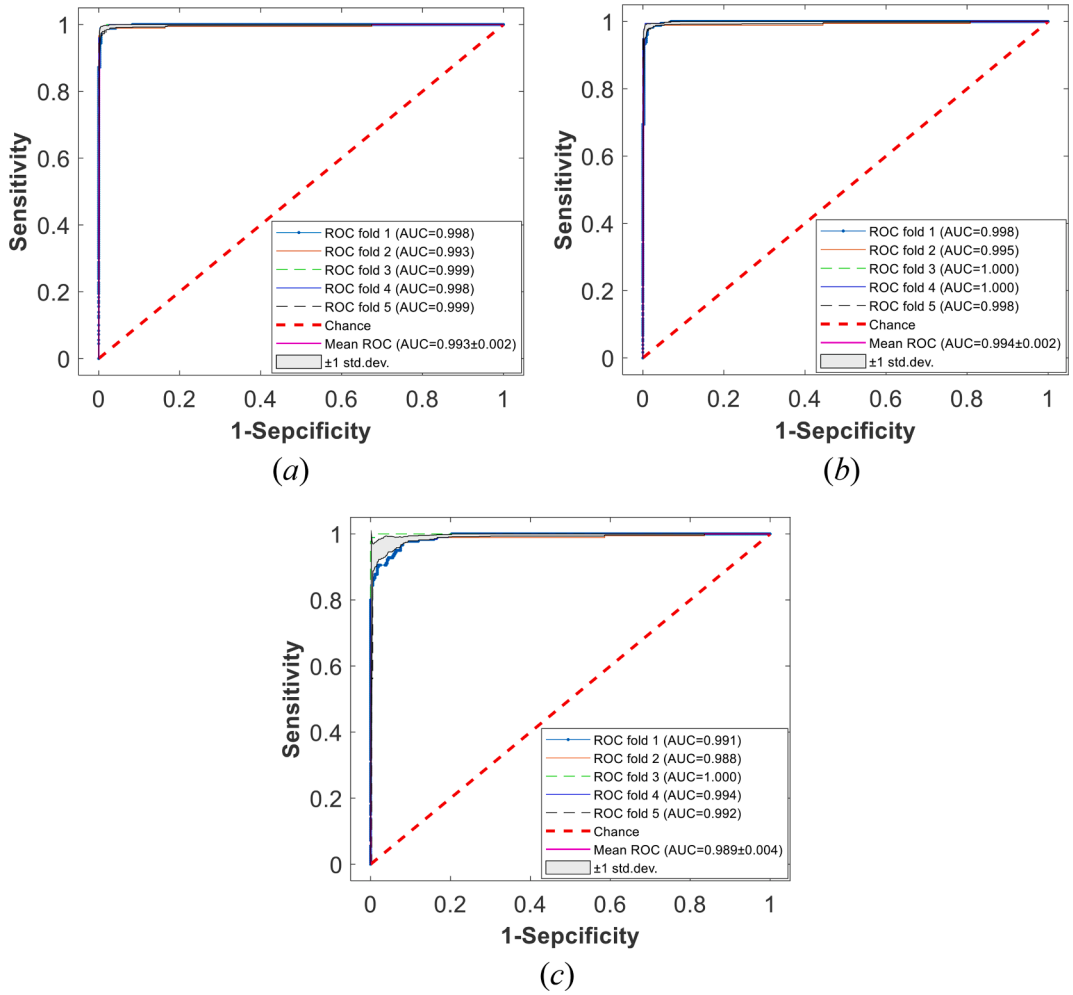


Fig. 9. ROC curves and AUCs obtained using 5-fold cross validation for the (*a*) original, (*b*) threshold, and (*c*) binary time Doppler spectrograms.

time–frequency features for fall activity detection [10]. The activity data of 4 actors were used to verify the classification performance, with the highest accuracy being 93% [10]. Branka Jokanović et al. adopted two stacked autoencoders to auto-extract features and used a logistic regression classifier to detect falls [22]. The highest accuracy was 97.1%, with a sensitivity of 79.0% when using a dataset, including 117

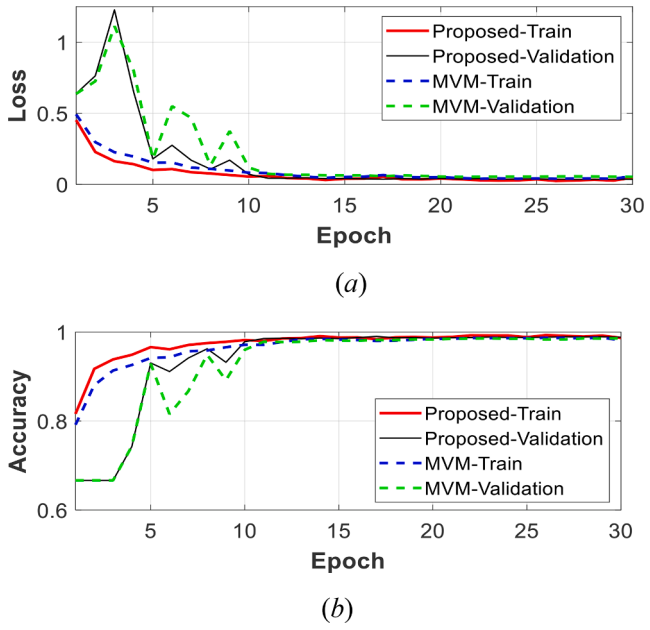


Fig. 10. (*a*) Cross-entropy loss and (*b*) accuracy values for the training and validation datasets using the proposed method and the MVM.

Table 4
Classification metrics obtained using the proposed method and the MVM using 5-fold cross-validation in Scene 1.

Metrics (%)	Method	Acc.	Pre.	Sen.	Spe.	$F_{1\text{-score}}$
Proposed	Original	98.926	98.443	98.333	99.222	98.388
	Threshold	98.778	98.328	98.000	99.167	98.164
	Binary	96.667	93.455	96.778	96.611	95.087
MVM	Original	98.630	97.262	98.667	98.611	97.959
	Threshold	96.556	93.063	96.889	96.389	94.937
	Binary	92.815	87.957	90.889	93.778	89.399

Abbreviations: Acc., Accuracy; Pre., Precision; Sen., Sensitivity; Spe., Specificity.

Table 5
Classification metrics of testing in Scene 2 using the proposed method and the MVM.

Metrics (%)	Method	Acc.	Pre.	Sen.	Spe.	$F_{1\text{-score}}$
N Proposed	Original	**95.370**	**94.286**	**91.667**	**97.222**	**92.958**
	Threshold	92.593	83.981	96.111	90.833	89.637
	Binary	92.778	87.701	91.111	93.611	89.373
MVM	Original	93.333	96.753	82.778	98.611	89.222
	Threshold	91.111	89.286	83.333	95.000	86.207
	Binary	88.519	86.875	77.222	94.167	81.765

Abbreviations: Acc., Accuracy; Pre., Precision; Sen., Sensitivity; Spe., Specificity.

fall activities and 291 non-fall activities from 3 subjects [22]. Baris Erol et al. proposed multidimensional principal component analysis and three-dimensional joint-variable radar signal representation, which could achieve 97.88% accuracy when using a dataset from 6 subjects [23]. Fall detection in an indoor environment using UWB radars and a CNN-LSTM architecture was realised with the best accuracy of 98.51% at one position using the leave-one-subject-out (total 10 male subjects) strategy [24]. Hamidreza Sadreazami et al. proposed a fall detection method using ultrawideband radar and a CNN that relied on multilevel feature learning from radar time–frequency representations [14]. The 5-fold cross validation in one scene achieved 93.54% accuracy using data from 10 subjects [14]. However, the accuracy decreased to 89.12% in another test scene [14]. At present, our radar can only detect falls of the human body in one room due to the limitation of radar numbers and transmission radio frequency. Injurious fall events cannot yet be distinguished from noninjury-related falls. This work has not researched the real-time monitoring of falls over a long period of time and it has not distinguished falls between humans and animals. Due to ethical constraints, the data on falls of the elderly were not allowed to be collected in this experiment with risk. In the future, we will try to combine multiple radars or implement a through-wall frequency band to detect falls in multiple rooms. The data in nursing homes, including the elderly and animals at the same time, should be collected to truly realise effective and robust fall monitoring and notification. Identifying injuries that fall from noninjury-related falls is worth studying.

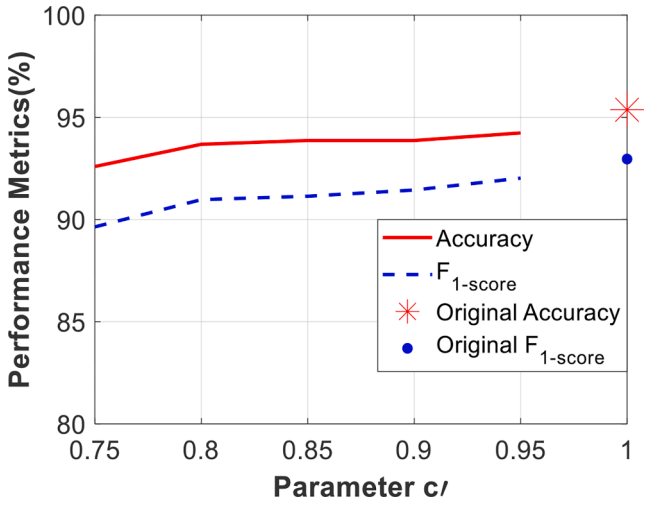


Fig. 11. The accuracy and the $F_{1\text{-score}}$ of testing in Scene 2 vary with parameter c'. The asterisk and the dot are the accuracy and the $F_{1\text{-score}}$ values using the original time Doppler spectrograms.

5. Conclusion

In this work, a novel optimum target range bin selection method for time–frequency analysis to detect falls is proposed based on wideband radar and the MobileNetV3-Small network. Data from 36 subjects were collected in two real room environments without any constraints, including 10 fall and 20 non-fall activities. The proposed optimum target range bin selection method based on the range Doppler spectrograms outperforms the MVM in both 5-fold cross validation and testing. The input of the original and the threshold time Doppler spectrograms to the deep learning network achieves a clearly better classification performance than that of the binary time Doppler spectrograms. The higher classification metrics indicate the robustness and the generalisability of the proposed method, even in unseen scenes and for unseen subjects.

CRediT authorship contribution statement

Mi He: Conceptualization, Methodology, Software, Formal analysis, Writing (original draft, review & editing), Visualization, Supervision, Project administration and Funding acquisition. **Yi Yang:** Software, Methodology, Validation, Formal analysis, Visualization. **Qinwen Ping:** Software, Validation, Formal analysis. **Ran Dai:** Investigation, Data curation. **Bingwen Liu:** Investigation, Resources. **Yongjian Nian:** Writing-review and Funding acquisition. **Zhu Zhang:** Writing-review.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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